

CURIONEST

Chapter 1 — Essential Ideas

Unit Pack · Lab Safety · Properties of Matter · Measurement · Sig Figs · Dimensional Analysis

WHAT'S INSIDE

- ✓ Complete concept notes with worked examples
- ✓ Practice sets in varied formats (short answer, MCQ, calculations)
- ✓ Key terms glossary + full answer key with worked solutions
- ✓ Original figures and Word-native equations — print-ready & editable

Made by a chemistry teacher — aligned to the standard US chemistry sequence.

In this unit, you will learn to:

1. Read the NFPA 704 hazard diamond and identify chemical hazards
2. Distinguish physical properties from chemical properties
3. Use SI units and metric prefixes correctly
4. Count significant figures and report calculations correctly
5. Convert units using dimensional analysis

Unit Vocabulary — Write each term in your own words

hazard diamond (NFPA 704)

physical property

chemical property

extensive property

intensive property

SI unit

metric prefix

significant figure

accuracy

precision

dimensional analysis

density

Lesson 1 — Lab Safety: The NFPA 704 Hazard Diamond

Objective: Read the four sections of the NFPA hazard diamond and explain what each rating means.

Before you touch any chemical, read its hazard information. The NFPA 704 hazard diamond — sometimes called a 'fire diamond' — summarizes the major hazards of a chemical in one symbol. It has four colored sections: the top (red) diamond specifies the level of fire hazard, based on the flash point temperature. The blue (left) diamond indicates the health hazard. The yellow (right) diamond describes reactivity — how readily the substance will detonate or undergo violent chemical change. The white (bottom) diamond points out special hazards, such as oxidizers or substances that react with water.

Each hazard is rated from 0 to 4: 0 means no hazard, 4 means extremely hazardous. A chemical rated 4 for health is deadly even with brief exposure; a rating of 1 means mild irritation. Reading these numbers tells you how careful you need to be.

Worked Example

A container shows a hazard diamond rated 2-3-1 with a W in the white section. What do the ratings tell you?

Step 1: Fire hazard (top, red) = 2: the substance must be moderately heated before it will ignite.

Step 2: Health hazard (left, blue) = 3: even short exposure can cause serious injury or permanent damage.

Step 3: Reactivity (right, yellow) = 1: stable normally, but can become unstable under high heat or pressure.

Step 4: Special (bottom, white) = W: the substance reacts violently with water.

Answer: Handle with full PPE, away from water and flames.

Check Your Understanding

1. Which section of the NFPA diamond indicates fire hazard?

- (A) Top (red)
- (B) Left (blue)
- (C) Right (yellow)
- (D) Bottom (white)

Answer: _____

2. A rating of 4 in any section means the hazard is:

- (A) No hazard
- (B) Slight
- (C) Serious
- (D) Extremely hazardous

Answer: _____

3. What does the W in the white section of the diamond mean?

- (A) Wet-able substance
- (B) Reacts violently with water
- (C) Water-soluble
- (D) Weigh carefully

Answer: _____

4. A chemical is rated 0 for fire hazard. What does this mean?

- (A) It will not burn
- (B) It burns easily
- (C) It explodes in air
- (D) It has a low flash point

Answer: _____